Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 10 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 3 : Session â€“ 10 : SLO â€“ 1 | Implementation of Queue Using Array â€” Code

I. Declaration and Initialisation

- #define MAX 100
- int queue[MAX];
- int front = -1, rear = -1;

II. Enqueue Function

- void enqueue(int val) {
- if (rear == MAX - 1) { printf("OVERFLOW\n"); return; }
- if (front == -1) front = 0;
- queue[++rear] = val;
- }

III. Dequeue Function

- int dequeue() {
- if (front == -1 || front > rear) { printf("UNDERFLOW\n"); return -1; }
- int val = queue[front++];
- if (front > rear) { front = rear = -1; }
- return val;
- }